

SDEV 220 - Individual Project Propos

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Project Title: BrewTrack: Coffee Shop Inventory Management System

1. Name of the Organization

Bean & Brew Co. (Local Independent Coffee Roastery and Café)

2. Description of the Problem/Need

Currently, Bean & Brew Co. relies on manual paper logs and spreadsheets to track their high-turnover inventory, such as specialty beans, dairy, and syrups. This leads to several critical issues:

- **Inaccurate Stock Levels:** Human error in manual counting leads to unexpected shortages of key items during peak hours.
- **Wasted Revenue:** Perishable items (like milk) expire before use because there is no automated tracking of "First-In, First-Out" or expiration dates.
- **Supplier Confusion:** Reordering is reactive rather than proactive, often resulting in rush shipping fees from multiple different suppliers.

3. Summary of Proposed System

I propose to build **BrewTrack**, a Python-based inventory management system designed specifically for coffee shop staff. The system will automate the relationship between inventory levels, supplier orders, and usage reports. By centralizing data into structured classes like `InventoryItem` and `Supplier`, the system will provide a real-time snapshot of the shop's assets and financial health.

4. Main Features and Functionality

Based on the provided UML design, the system will include the following functionality:

- **Dynamic Inventory Tracking:** Update stock levels instantly via the `updateStock()` method, allowing staff to add new shipments or deduct daily usage.

- **Automated Reorder Alerts:** The `checkReorderLevel()` function will flag items that fall below a specific threshold (e.g., Espresso Beans < 10 bags) to prevent shortages.
- **Integrated Supplier Management:** A dedicated `Supplier` class to store contact details and track which items are "suppliedBy" specific vendors for streamlined ordering.
- **Order Lifecycle Management:** Functionality to create, process, and track the status of supply orders (`Pending`, `Shipped`, `Delivered`) to ensure the supply chain is transparent.
- **Business Intelligence Reporting:** A `ReportGenerator` to produce inventory valuation and usage reports, helping the owner understand cost of goods sold (COGS).
- **Secure Staff Access:** A `User` login system to ensure only authorized staff can update inventory or view sensitive financial reports.

5. Explanation of Requirements Fulfillment

This project meets the SDEV 220 requirements by:

- **Object-Oriented Programming (OOP):** Using at least six distinct classes with inheritance and associations as outlined in the UML diagram.
- **Data Validation:** Implementing logic to ensure stock cannot be negative and that order IDs stay unique.
- **Collaboration Readiness:** Though an individual project, the architecture follows industry-standard UML patterns, making it "collaboration-ready" for a GitHub environment.
- **Practical Application:** The system addresses a real-world business need using Python logic to solve a planning problem.